shown in Fig. 13. The GPIO pins defined in this project for switching the four relays are shown in Fig. 14.

To control four appliances, the author used GPIO2, GPIO16, GPIO4, and GPIO5 pins of NodeMCU. Each GPIO pin should be connected to corresponding relay module, as shown in connection diagram of Fig. 6.

Next is “cha_switch_on.setter = cha_switch_on_setter;” line of code. Copy this one too and add it as many times as the number of devices to be added in the project. For adding four new devices, change the switch_on and setter, as shown in Fig. 15.

Next is “void my_homekit_setup()” line. Inside that you will find “pinMode(PIN_SWITCH, OUTPUT);

digitalWrite(PIN_SWITCH, HIGH);” You have to add these lines as many times as you have added new devices. For four devices, you need to change the variables, as shown in Fig. 16.

After this select following lines: “void cha_switch_on_setter

```
void my_homekit_setup() {
  pinMode(PIN_SWITCH, OUTPUT);
  digitalWrite(PIN_SWITCH, HIGH);

  pinMode(PIN_SWITCH2, OUTPUT);
  digitalWrite(PIN_SWITCH2, HIGH);

    pinMode(PIN_SWITCH3, OUTPUT);
  digitalWrite(PIN_SWITCH3, HIGH);

    pinMode(PIN_SWITCH4, OUTPUT);
  digitalWrite(PIN_SWITCH4, HIGH);
```

Fig. 16: PinMode variables

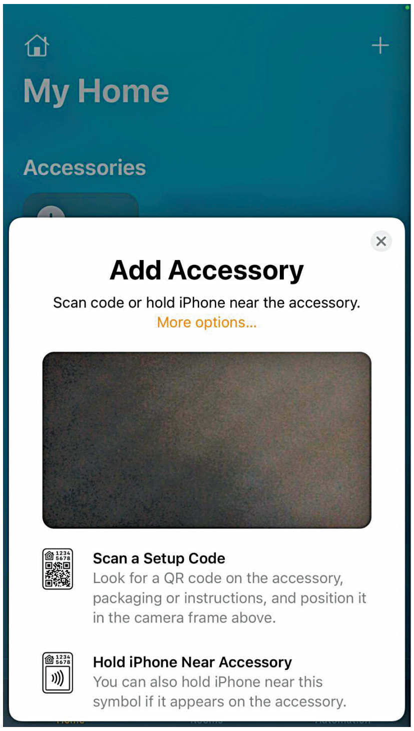

Fig. 18: Add Accessary window

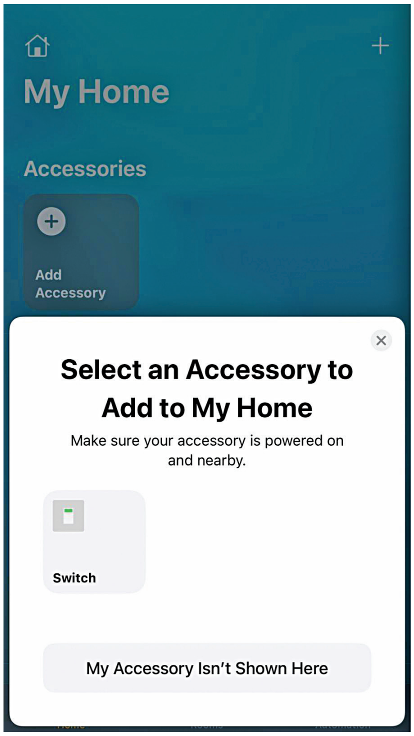

Fig. 19: Add To My Home window

```
void cha_switch_on_setter(const homekit_value_t value) {
  bool on = value.bool_value;
  cha_switch_on.value.bool_value = on;  //sync the value
  LOG_D("Switch: %s", on ? "ON" : "OFF");
  digitalWrite(PIN_SWITCH, on ? LOW : HIGH);
}

void cha_switch_on_setter2(const homekit_value_t value) {
  bool on = value.bool_value;
  cha_switch_on2.value.bool_value = on;  //sync the value
  LOG_D("Switch: %s", on ? "ON" : "OFF");
  digitalWrite(PIN_SWITCH2, on ? LOW : HIGH);
}

void cha_switch_on_setter3(const homekit_value_t value) {
  bool on = value.bool_value;
  cha_switch_on3.value.bool_value = on;  //sync the value
  LOG_D("Switch: %s", on ? "ON" : "OFF");
  digitalWrite(PIN_SWITCH3, on ? LOW : HIGH);
}

void cha_switch_on_setter4(const homekit_value_t value) {
  bool on = value.bool_value;
  cha_switch_on4.value.bool_value = on;  //sync the value
  LOG_D("Switch: %s", on ? "ON" : "OFF");
  digitalWrite(PIN_SWITCH4, on ? LOW : HIGH);
}
```

Fig. 17: The .setter function to get the switch-events

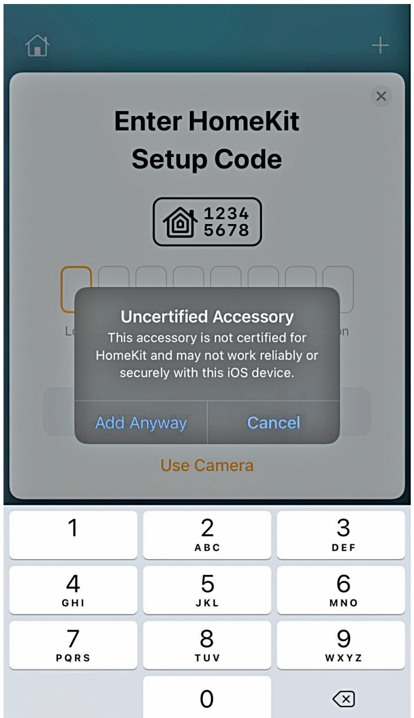

Fig. 20: Setup warning message